Kamp's Theorem

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1 Statement and definitions

This is based off of Rabinovich [2014]

Definition 1 (**LTL**). *Let M_w be a word-structure over alphabet Σ and $i \in [n]$ for $n = |M_w|$ be a position in the word.*

$$
\begin{array}{ll}
M_w, i \models Q_a & \text{if } w_i = a \text{ for } a \in \Sigma \\
M_w, i \models \phi_1 \vee \phi_2 & \text{if } M_w, i \models \phi_1 \text{ or } M_w i \models \phi_2 \\
M_w, i \models \neg\phi_1 & \text{if } M_w i \not\models \phi_1 \\
M_w, i \models \phi_1 \textbf{ since } \phi_2 & \text{if for some } j < i, \text{ we have } M_w j \models \phi_2, \\
 & \text{and for all } k \text{ such that } j < k < i, \text{ we have } M_w, k \models \phi_1 \\
M_w, i \models \phi_1 \textbf{ until } \phi_2 & \text{if for some } j > i, \text{ we have } M_w, j \models \phi_2, \\
 & \text{and for all } k \text{ such that } i < k < j, \text{ we have } M_w, k \models \phi_1.
\end{array}
$$

We use abbreviations $\mathbf{K}^+(F) := \neg((\neg F) \textbf{ until } \top)$ and $\mathbf{K}^-(F) := \neg((\neg F) \textbf{ since } \top)$

$$
M_w, t \models \mathbf{K}^-(F) \iff t = \sup(\{t' \mid t' < t \text{ and } M_w, t' \models F\})
$$
$$
M_w, t \models \mathbf{K}^+(F) \iff t = \inf(\{t' \mid t' > t \text{ and } M_w, t' \models F\})
$$

Theorem 2. *Given any $\boldsymbol{FO}[<]$ formula $\varphi(x)$ over one free variable, there is an **LTL** formula F which is equivalent to φ over Dedekind complete chains (if a non-empty subset has a lower (upper) bound, it has a greatest lower (least upper) bound). That is, for any M_w and position i in M_w:*

$$
M_w \models \varphi(i) \iff M_w, i \models F
$$

Proof. Sketch: Every $\varphi(x)$ is equivalent to a disjunction of $\overrightarrow{\exists}\forall$ formulas (defined in next section) with one free variable over Dedekind complete chains. Then, every $\overrightarrow{\exists}\forall$ formula with one free variable is equivalent to an **LTL** formula. $\square$

2 Preliminaries

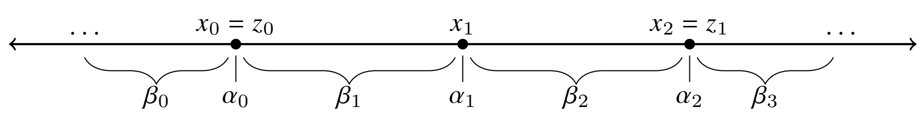
2.1 $\overrightarrow{\exists}\forall$ Formulas

Definition 3 ($\overrightarrow{\exists}\forall$ formula)**.** *Here α_i, β_i will denote temporal subformulas. This construction is related to Morleyization in model theory. Let α_j and β_j be monadic predicates. A $\overrightarrow{\exists}\forall$ formula, called a "decomposition formula" is a formula of the form*

$$
\begin{aligned}
\psi(z_0, \ldots, z_m) := \exists x_n \ldots \exists x_1 \exists x_0 \\
& \left(\bigwedge_{k=0}^{m} z_k = x_{i_k} \right) \wedge (x_n > x_{n-1} > \ldots > x_1 > x_0) \\
& \wedge \bigwedge_{j=0}^{n} \alpha_j(x_j) \\
& \wedge \bigwedge_{j=0}^{n} \left((\forall y)^{<x_j}_{>x_{j-1}} \beta_j(y) \right) \\
& \wedge (\forall y)^{<x_0} \beta_0(y) \\
& \wedge (\forall y)_{>x_{n+1}} \beta_{n+1}(y)
\end{aligned}
$$

A disjunction of $\overrightarrow{\exists}\forall$ formulas is notated $\vee\overrightarrow{\exists}\forall$.

Essentially, a $\overrightarrow{\exists}\forall$ specifies n many ordered events on a timeline and what happens inbetween those events.



Proposition 4. *Every $\vee\overrightarrow{\exists}\forall$ formula with one free variable is equivalent to an **LTL** formula*

Proof. Let $\Box(F) := \neg(\top$ **until** $(\neg F))$ abbreviate "henceforth F" and $\overleftarrow{\Box}(F) :=$ $\neg(\top$ **since** $(\neg F))$ similarly. Observe if we have a $\overrightarrow{\exists}\forall$ formula $\psi(z_0)$ with one free variable z_0 that is set to x_k, and corresponding temporal formulas A_i, B_i for each α,β_i, then ψ is equivalent to the conjunction of

$$A_k \wedge (B_{k+1} \textbf{ until } (A_{k+1} \wedge (B_{k+2} \textbf{ until } \ldots (A_{n-1} \wedge (B_n \textbf{ until } (A_n \wedge \Box B_{n+1})) \ldots))))$$

and

$$A_k \wedge (B_{k-1} \textbf{ since } (A_{k-1} \wedge (B_{k-2} \textbf{ since } (\ldots A_1 \wedge (B_1 \textbf{ since } (A_0 \wedge \overleftarrow{\Box} B_0)) \ldots))$$

$\Box$

2.2 Closure properties

Lemma 5. *Every $\overrightarrow{\exists}\forall$ formula is equivalent to the conjunction of $\overrightarrow{\exists}\forall$ formulas with at most two free variables.*

Lemma 6. *The set of $\vee\overrightarrow{\exists}\forall$ formulas is closed under $\vee$, $\wedge$, and $\exists$.*

Negation is not straightforward.

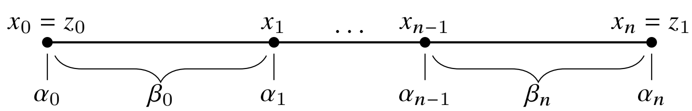
3 Negation of $\overrightarrow{\exists}\forall$ formulas

Proposition 7. *The negation of $\overrightarrow{\exists}\forall$ formulas with at most two variables is equivalent over Dedekind complete chains to a disjunction of $\overrightarrow{\exists}\forall$ formulas.*

Let $\psi(z_0, z_1)$ be such a $\overrightarrow{\exists}\forall$ formula. Intuitively:

$$\begin{aligned}\psi(z_0, z_1) \equiv\ & \text{specification of events before } z_0\\ & \wedge \text{specification of events between } z_0, z_1\\ & \wedge \text{specification of events after } z_1\end{aligned}$$

The first and third only depend on 1 variable, so they are immediately convertable into monadic **LTL** formulas. So in the second, we can restrict the timeline we look at to a finite interval.



So we only really need to consider this case. That is,

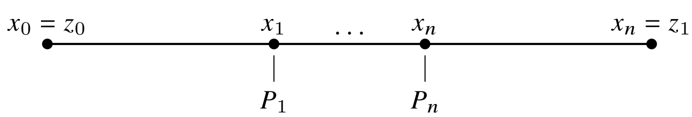
Proposition 8. *The negation of any formula of the form*

$$\exists x_0 \ldots \exists x_n \left((z_0 = x_0 < \ldots < x_n = z_1) \wedge \bigwedge_{j=0}^{n} \alpha_j(x_j) \wedge \bigwedge_{j=1}^{n} (\forall y)_{>x_j-1}^{<x_j} \beta_j(y) \right)$$

where α_i, β_i are quantifier free, is equivalent over Dedekind complete chains to a disjunction of $\overrightarrow{\exists}\forall$ formulas.

Lemma 9. $\neg\exists x_1 \ldots \exists x_n (z_0 < x_1 < \ldots x_n < z_1) \wedge \bigwedge_{i=1}^{n} P_i(x)$ *is equivalent over Dedekind complete chains to a $\vee\overrightarrow{\exists}\forall$ formula $O_n(P_1, \ldots, P_n, z_0, z_1)$*

Proof. So we don't specify the endpoints or the intervals between any points.



We will induct on the number of events n. If $n = 1$, then $\neg(\exists x_1)^{<z_1}_{>z_0} P_1(x_1)$ is equivalent to $(\forall y)^{<z_1}_{>z_0} \neg P_1(y)$, a $\overrightarrow{\exists}\forall$ formula. Otherwise assume O_n exists and we can construct a $\vee\overrightarrow{\exists}\forall$ formula O_{n+1}. If (z_0, z_1) is non-empty:

- If P_1 does not occur in (z_0, z_1), then $(\forall y)^{<z_1}_{>z_0} \neg P_1(y)$ and $O_{n+1} \equiv \top$
- If P_1 does occur in (z_0, z_1), then $r_0 = \inf(\{z \in (z_0, z_1) \mid P_1(z)\})$ exists by Dedekind completeness. And we can detect it using the following formula

  $$INF(z_0, r_0, z_1, P_1) := (z_1 < r_0 < z_1) \wedge (\forall y)^{<r_0}_{>z_0} \neg P_1(y) \wedge (P_1(r_0) \vee \mathbf{K}^+(P_1)(r_0))$$

  There are two subcases

  – If $r_0 = z_0$, then $O_{n+1}(P_1, \ldots, P_{n+1}, z_0, z_1) := O_n(P_2, \ldots, P_n, z_0, z_1)$
  – If $r_0 \in (z_0, z_1)$, then $O_{n+1}(P_1, \ldots, P_{n+1}, z_0, z_1) := O_n(P_2, \ldots, P_n, r_0, z_1)$

Thus $O_{n+1}(P_1, \ldots, P_{n+1}, z_0, z_1)$ can be defined as the disjunction of

- $(\forall y)^{<z_1}_{>z_0} \bot$
- $(\forall y)^{<z_1}_{>z_0} \neg P_1(y)$
- $\mathbf{K}^+(P_1)(z_0) \wedge O_n(P_2, \ldots, P_n, z_0, z_1)$
- $(\exists r_0)^{<z_1}_{>z_0}(INF(z_0, r_0, z_1, P_1) \wedge O_n(P_2, \ldots, P_n, r_0, z_1))$

$\mathbf{K}^+(P_1)$ is atomic, INF is $\vee\overrightarrow{\exists}\forall$, so byt closure properties O_{n+1} is $\vee\overrightarrow{\exists}\forall$ □

3.1 Proof of Proposition 8

We want to show the negation of

$$\exists x_0 \ldots \exists x_n \left((z_0 = x_0 < \ldots < x_n = z_1) \wedge \bigwedge_{j=0}^{n} \alpha_j(x_j) \wedge \bigwedge_{j=1}^{n} (\forall y)^{<x_j}_{>x_j - 1} \beta_j(y) \right)$$

is a $\vee\overrightarrow{\exists}\forall$ formula. Just a sketch for now:

Proof. We divide into cases again

1. $\neg\alpha_0(z_0)$ or $\mathbf{K}^+(\neg\beta_1)(z_0)$ (immediate)
2. $\alpha_0(z_0)$ and β_1 holds all along (z_0, z_1) (use a corollary of the previous lemma)
3. Both
   (a) $\alpha(z_0) \wedge \neg\mathbf{K}^+(\neg\beta_1)(z_0)$
   (b) there is $x \in (z_0, z_1)$ such that $\neg\beta_1(x)$

   Requires care, but the idea is to locate the first occurrence of $\neg\beta_1$, and we can place x_1 before it and then work from there.

□

References

Alexander Rabinovich. A proof of kamp's theorem. *Logical Methods in Computer Science*, 10, 2014.